

Home Credit - BI Analyst Test Solutions

May 12, 2025

Introduction

This document is my solution to the Home Credit BI Analyst Test, prepared as a guide for students to learn and understand SQL, logical reasoning, and analytical skills. I hope this detailed explanation will help you grasp the concepts effectively and inspire confidence in your studies.

I believe that this task is not an overly difficult challenge for you, and I hope that after referring to the solution section, you have confidently completed at least 80% of the task. If your results have not met your expectations, do not be discouraged! This is an opportunity for you to reflect, invest more time in practice, and hone your skills. Every small step forward today lays a solid foundation for great successes tomorrow. I hope that you will maintain a strong spirit of learning, continuously strive, and persistently pursue your goals. I am confident that with dedication and determination, you will achieve many outstanding accomplishments in the future—not only in this subject but also on your journey to conquer knowledge ahead. Be confident, for you are fully capable of succeeding!

Keep working hard, stay curious, and believe in your potential—you can achieve great results with dedication and effort!

1 Section A: SQL Test (4 Points)

1.1 Question 1: Detailed Information About All Credits and Clients (0.5 Points)

Question: Which query returns detailed information about all credits and clients who got those credits?

- a) `select * from view_credit c right join view_person p on p.id_person = c.id_person`
- b) `select * from view_credit c join view_person p on p.id_person = c.id_person`
- c) `select * from view_credit c left join view_person p on p.id_person = c.id_person`
- d) `select * from view_credit c full outer join view_person p on p.id_person = c.id_person`

Answer: The correct answer is (c).

Explanation: To retrieve detailed information about all credits and the clients who obtained them, we need to start with the `view_credit` table (which contains all credits) and join it with the `view_person` table (which contains client information). A `LEFT JOIN` ensures that all records from `view_credit` are included, even if there are no matching records in `view_person`. The join condition `p.id_person = c.id_person` links the two tables. Options (a) and (d) might include unnecessary records, and (b) (an inner join) would exclude credits without matching clients.

1.2 Question 2: Detailed Information About All Clients and Their Credits (0.5 Points)

Question: Which query returns detailed information about all clients and credits they got (i.e., it shows all clients and credits data for those who have it)?

- a) `select * from view_person p join view_credit c on p.id_person = c.id_person`
- b) `select * from view_credit c join view_person p on p.id_person = c.id_person`
- c) `select * from view_credit c left join view_person p on p.id_person = c.id_person`
- d) `select * from view_person p left join view_credit c on p.id_person = c.id_person`

Answer: The correct answer is (d).

Explanation: To show all clients (from `view_person`) and their credits (from `view_credit`), we need a `LEFT JOIN` or `FULL OUTER JOIN` starting from `view_person`. A `RIGHT JOIN` would ensure all records from `view_person` are included, along with matching records from `view_credit`. Option (a) uses an inner join, which only includes clients with credits. Option (b) is also an inner join, and (c) starts from `view_credit`, which is incorrect for this question. The correct query should be: d

1.3 Question 3: Clients Without Any Credit (0.5 Points)

Question: Which query returns information about all clients who don't have any credit?

- a) `select c.* from view_person p left join view_credit c on p.id_person = c.id_person where c.id_person is null`
- b) `select p.* from view_person p left join view_credit c on p.id_person = c.id_person where c.id is null`
- c) `select p.* from view_person p where p.id_person not in (select p.id_person from view_credit)`
- d) `select p.* from view_person p where p.id_person not in (select c.id_person from view_credit c where p.id_person = c.id_person)`

Answer: The correct answer is (b).

Explanation: To identify clients who do not have any credits, we need to find records in `view_person` that have no corresponding entries in `view_credit`. Option (b) uses a `LEFT JOIN` from `view_person` to `view_credit`, ensuring all clients are included, and the condition `where c.id is null` filters for clients with no matching credit records, as `c.id` is the primary identifier in `view_credit`. Option (a) incorrectly selects from `view_credit` and checks `c.id_person is null`, which is not appropriate since `id_person` may not be null in the join context. Option (c) has a syntax error in the subquery, referencing `p.id_person` instead of `c.id_person`, and is less efficient. Option (d) includes an invalid subquery condition `where p.id_person = c.id_person`, which would always return an empty result set due to the correlation issue. Thus, (b) is the correct query.

1.4 Question 4: Clients With More Than 1 Credit (0.5 Points)

Question: Which query returns information about all clients who have more than 1 credit?

- a) `select p.* from view_person p where (select count(c.id) from view_credit c where c.id_person=p.id_person)>1`
- b) `select p.* from view_person p where (select count(c.id) from view_credit c where c.id_person=p.id_person)=2`
- c) `select p.* from view_person p join (select count(*) k, c.id_person from view_credit c group by c.id_person having count(*) > 1) g on p.id_person=g.id_person`
- d) `select p.* from view_person p join (select count(*) k, c.id_person from view_credit c group by c.id_person where count(*) > 1) g on p.id_person=g.id_person`

Answer: The correct answer is (a).

Explanation: To find clients with more than 1 credit, we need to count the number of credits per client and filter for those with a count greater than 1. Option (a) uses a subquery to count credits per `id_person` and correctly filters with `>1`, returning all clients with more than 1 credit. Option (b) restricts the count to exactly 2, missing clients with 3 or more credits. Option (c) has a syntax error: `group by c.id` is incorrect since `c.id` is the credit ID, not the person ID; it should be `group by c.id_person`. Option (d) also has a syntax error: `where count(*) > 1` is not valid in this context; it should use `HAVING` instead of `WHERE` after `GROUP BY`. Thus, (a) is the correct and most straightforward query.

1.5 Question 5: What query returns this result (1 Points)

Question: Write SQL

Answer:

```
SELECT c.INTEREST, SUM(CREDIT_AMOUNT) AS CREDIT_AMOUNT_SUM,  
MAX(CREDIT_AMOUNT) AS CREDIT_AMOUNT_MAX, COUNT(*) AS Accs  
FROM view_credit c  
WHERE c.INTEREST > 0.18  
ORDER BY c.INTEREST ASC
```

1.6 Question 6: Average Age of Clients with Credits (0.5 Points)

Question: Which query returns the average age of clients who have credits?

- a) `select avg((sysdate-p.date_birth)/365.25) from view_person p`
- b) `select avg((sysdate-p.date_birth)/365.25) from view_person p where p.id_person in (select id_person from view_credit c)`
- c) `select avg((sysdate-p.date_birth)/365) from view_person p left join view_credit c on p.id_person = c.id_person`
- d) `select avg((sysdate-p.date_birth)/365) from view_person p`

Answer: The correct answer is (b).

Explanation: To calculate the average age of clients who have credits, we need to filter `view_person` to include only those clients who have records in `view_credit`. Option (b) uses a subquery to achieve this: `where p.id_person in (select id_person from view_credit c)`. The age is calculated by subtracting the birth date from the current date (`sysdate`) and dividing by 365.25 (accounting for leap years). Option (a) includes all clients, not just those with credits. Option (c) uses a `LEFT JOIN` without filtering, which may include clients without credits. Option (d) also includes all clients.

1.7 Question 7: Average Number of Children for All Clients (0.5 Points)

Question: Which query returns the average number of children for all clients?

- a) `select avg(p.child_num) from view_person p`
- b) `select avg(p.child_num) from view_person p where p.id_person in (select id_person from view_credit c)`
- c) `select sum(p.child_num)/count(*) from view_person p where p.id_person in (select id_person from view_credit c)`
- d) `select avg(p.child_num) from view_person p where p.id_person in (select id_person from view_credit c where c.id_person=p.id_person)`

Answer: The correct answer is (a).

Explanation: The question asks for the average number of children for *all clients*, not just those with credits. Option (a) correctly calculates this by using `avg(p.child_num)`

across all records in `view_person`. Options (b), (c), and (d) incorrectly filter for clients with credits, which is not required by the question. Note that `avg(p.child_num)` is equivalent to `sum(p.child_num)/count(*)`, but (a) is the simplest and most direct approach.

2 Section B: Logical and Analytical Test (6 Points)

2.1 Question 1: Average Monthly Salary Cost (1 Point)

Question: There are 6 employees in a company. Two of them have a 11,000 USD yearly salary. One of them gets 8,000 USD per year. The others get 14,000 USD. What is the average monthly cost for salary for one employee in this company? Why?

Answer: The average monthly cost for salary per employee is 1,000 USD.

Explanation: First, calculate the total yearly salary for all 6 employees:

$$(11,000 \times 2) + 8,000 + (14,000 \times 3) = 22,000 + 8,000 + 42,000 = 72,000 \text{ USD}$$

Next, divide by 12 months to get the total monthly salary:

$$\frac{72,000}{12} = 6,000 \text{ USD}$$

Finally, divide by 6 employees to find the average monthly cost per employee:

$$\frac{6,000}{6} = 1,000 \text{ USD}$$

2.2 Question 2: Logical Statements (1 Point)

Question: Statements: All flowers are cats. All fans are cats.

Conclusions:

- I. All flowers are fans.
- II. Some fans are flowers.

- a) Only conclusion I follows
- b) Only conclusion II follows
- c) Either I or II follows
- d) Neither I nor II follows
- e) Both I and II follow

Answer: The correct answer is (d).

Explanation: Consider the statements: all flowers are cats, and all fans are cats. This means both flowers and fans are subsets of cats. However, there is no information indicating that flowers and fans overlap. For example, flowers and fans could be entirely separate groups within cats. Therefore: - Conclusion I (All flowers are fans) does not follow, as flowers and fans may not overlap. - Conclusion II (Some fans are flowers) also does not follow, as there is no guarantee of overlap. Thus, neither conclusion follows, making (d) the correct answer.

2.3 Question 3: Mean and Median (1 Point)

Question: There are 2 groups of people: one with 10 males (average age 25) and one with 10 females (average age 20). There are twins within the groups—a male and a female. Which of the following conclusions are generally valid?

-) Median age of men is bigger than the median age of women. (true/false)
-) If we sort both groups by age and create 10 pairs (youngest man with youngest woman, etc.), there will be 5 or more pairs where the man is older than or the same age as the woman. (true/false)
-) Groups can be rearranged by moving people from one group to another so the average age increases in both groups. (true/false)

Answer and Explanation:

a) Median age of men is bigger than the median age of women: False

Consider the following example:

	Male	Female	Note
1	19	20	
2	19	20	
3	19	20	
4	19	20	
5	19	20	
6	19	20	
7	19	20	
8	19	20	
9	20	20	Twins
10	78	20	
Average	25	20	
Median	19	20	

The female group has all members at age 20, so the median is 20. The male group has 8 members at age 19, 1 at 20 (twin), and 1 at 78, making the average 25 but the median 19 (since the 5th and 6th values are 19). Thus, the median age of men (19) is less than the median age of women (20), so the statement is false.

b) Pairing by age: False

Using the same example:

	Male	Pair	Female	Note
1	19	~	20	
2	19	~	20	
3	19	~	20	
4	19	~	20	
5	19	~	20	
6	19	~	20	
7	19	~	20	
8	19	~	20	
9	20	~	20	Twins
10	78	~	20	
Average	25		20	

Pairing the youngest to oldest, only the 9th pair (20 vs. 20) and 10th pair (78 vs. 20) have the man older or the same age as the woman. This is only 2 pairs, not 5 or more, so the statement is false.

c) Rearranging to increase average age: True

Consider two cases:

Case 1: Rearrange while keeping 10 people per group

Swapping individuals between groups either keeps the average the same (e.g., swapping the twins, both aged 20) or increases one group's average while decreasing the other's (e.g., swapping a 78-year-old male with a 20-year-old female). Thus, it's not possible to increase both averages while keeping group sizes equal.

Case 2: Rearrange without maintaining group sizes

Consider this example:

	Male	Female
1	20	20
Twins		
2	21	20
3	21	20
4	21	20
5	21	20
6	21	20
7	21	20
8	21	20
9	62	20
10		20
11		21
Male swapped		
Average	25.44	20.09

Move a 21-year-old male to the female group, resulting in 9 males (average 25.44) and 11 females (average 20.09). Both averages are higher than the original (25 and 20), so the statement is true.

2.4 Question 4: Analytics 1 (1 Point)

Question: Given the table of volumes (Smartphone, Motorbike, Cash) over months:

MONTH	Volume Smartphone	Volume Motorbike	Volume Cash
2020-09	12,491	40,365	23,283
2020-08	10,493	32,785	22,536
2020-07	9,499	27,585	23,213
2020-06	8,940	26,113	23,551
2020-05	13,045	30,203	19,335
2020-04	9,624	23,142	18,594
2020-03	14,138	35,749	28,229
2020-02	14,697	41,249	26,363
2020-01	26,085	92,258	24,109
2019-12	17,576	64,367	30,586
2019-11	14,150	54,114	29,249
2019-10	13,565	53,708	29,531
2019-09	14,360	69,609	32,572
2019-08	8,395	26,228	23,003

a) Why is the volume in January significantly higher than other months?

Answer: January is the first month of the year, often associated with bonuses and New Year celebrations. Employees, especially office workers, may receive annual bonuses or festive allowances, leading to increased spending on items like smartphones and motorbikes to start the year anew. Additionally, retailers often launch promotional campaigns to attract customers, boosting sales volume in January compared to other months.

b) Why is the volume of Motorbikes in September normally higher than in August and October?

Answer: September marks the end of a quarter, a time when sales teams and motorbike dealerships push to meet quarterly KPIs. They often introduce promotions and discounts to boost sales. Additionally, September coincides with the start of the academic year, increasing demand for motorbikes among students. In contrast, August is mid-quarter, and October is the start of a new quarter, where the urgency to meet KPIs is less intense, leading to lower volumes.

c) Why does the trend of Cash volume in June and July go up while others go down?

Answer: June and July are summer months when students are on holiday. Parents may take children on trips, leading to higher cash usage. Meanwhile, volumes for smartphones and motorbikes are lower, as these purchases may have already been made earlier in the year (e.g., January or September) for commuting or other purposes.

2.5 Question 4: Analytics 2 (1 Point)

Question: The table below shows the Cash Loan Target Achievement of Team Leaders over the last 13 months (Note: The table is not fully provided in the original document, but the explanation refers to color coding: green for 100%, yellow for 80%–99%, and red for < 80%).

Explanation (Student's Note): The table tracks the target achievement of team leaders over 13 months. Green indicates achievement of 100% or more, yellow indicates 80%–99%, and red indicates below 80%.

a) From your point of view, who are the best and worst Team Leaders? Why?

Answer:

+ I consider Internal 12 to be the best Team Leader, as only the month of 2020 has a percentage lower than the average of the teams, while Internal 12 consistently exceeds the expected target (over 100% in 10 months, with the remaining 3 months also achieving over 90% of the target). Additionally, when compared to other teams, Internal 12 largely leads in meeting targets.

+ The worst Team Leader is Internal 7, as when compared to the monthly average percentage, no month for this team exceeds the average monthly percentage (meaning this team is dragging down the overall average target percentage of the teams). Furthermore, there are many months below the 80% target, with only one month achieving over 100%, which is still the lowest compared to other teams.

b) Is there any unusual result in the above table? Why?

Answer: Upon reviewing the data table:

+ An unusual result is observed in April, where all teams failed to achieve a target of 100% or more.

Explanation: April is a high-workload month (for students and teachers, April is exam season, with teachers working intensively to teach), reducing the likelihood of borrowing, as there is little need for loans in April.

+ In May, the targets increase, while June and July show a decline:

Explanation: Since April is a study and exam preparation month, May marks the end of the exam season, increasing the likelihood that parents will borrow money for family trips or gatherings in June and July, causing the targets to drop in those months.

+ Additionally, it is noted that in August, all teams failed to meet targets, with several teams even falling into the red category (below 80% target). However, in September, most teams achieved high targets with a rapid increase, with September recording the highest target achievement.

Explanation:

- The low target in August is due to a traditional belief that borrowing money during the "Ghost Month" (the 7th lunar month, from August 8, 2021, to September 6, 2021, falling within August) could lead to significant debt, negatively affecting financial luck, especially on the 1st day of the month (a major taboo day). This makes earning money difficult, business less smooth, with more misfortune than luck, potentially leading to losses or bankruptcy. As a result, customers avoid borrowing, causing many teams to miss their targets.

- The rapid increase in targets in September is because, after the Ghost Month ends, people resume borrowing to purchase goods (September being the end of the quarter, featuring many promotional items), naturally leading to a rise in targets.